ASEEM R. KSHIRSAGAR



Physicist | Computational Material Scientist

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EDUCATION

PhD in "*Ab initio* study of the excited state properties of photoresponsive metal-organic frameworks"

SIMAP laboratory, University of Grenoble-Alps

Nov 2017-Mar 2021 — Grenoble, France

Bachelor of Science-Master of Science (BS-MS) dual degree

Indian Institute of Science Education and Research (IISER)

Aug 2012- May 2017 — Pune, India

RESEARCH EXPERIENCE

MOPGA Fellow : Make Our Planet Great Again fellowship, Campus France and MEAE

"Stabilizing halide perovskites for efficient and durable solar cells"

Rennes Institute of Chemical Sciences, CNRS, University of Rennes

1 Dec. 2025 - Ongoing — Rennes, France

- Electronic structure description of novel 1D halide perovskites, including exciton-fine structure, carried out using many-body perturbation theory and group theoretical analysis
- Side project: literature review on exciton-polarons in polar semiconductors, and supervision of PhD student working on GW description of electronic structure of TlCl polar semiconductor

Advisor: Claudine Katan **Collaborators**: Jacky Even, Aditya Mohite

Postdoctoral Fellow : Brittany Regional Fellowship for Sustainability Research

"Optical properties of chiral metal-halide perovskites"

Rennes Institute of Chemical Sciences, CNRS, University of Rennes

1 Dec. 2024 - Ongoing — Rennes, France

- Theoretical description of lattice dynamics in chiral 1D hybrid metal halides with
- Computational description of circular dichroism in 2D chiral hybrid metal halide perovskites, tracing its dependence of crystal orientation and chemical substitution to underlying electronic structure
- Side project: *ab initio* description of electronic structure of novel lead-free metal halides relevant for building light-sensing devices
- Supervision of masters student working on computation of mechanical angular momentum of chiral phonons

Advisor: Claudine Katan **Collaborators**: Mikaël Kepenekian, Jacky Even

Postdoctoral Researcher

"Resonant Raman scattering in $BiFeO_3$"

Department of Physics and Material Science, University of Luxembourg

1 July 2021 - 30 Nov. 2024 — Luxembourg

- Study of spin-flip electronic excitations in $BiFeO_3$ using Bethe-Salpeter formalism and their possible use for light-induced spin-magnetization
- Tracing in-gap Raman resonance in $BiFeO_3$ to strong coupling of spin-flip electronic excitations with lattice vibrations
- Side projects: Raman scattering in $BaZrS_3$ and $BaHfS_3$, Ambipolar current-voltage behaviour of Ti-WSe_2 system

Advisor: Sven Reichardt **Collaborators**: Ludger Wirtz, Mäel Guennou, Georgy Gordeev

Doctoral Researcher

"Ab initio study of the excited state properties of azobenzene-functionalized photoresponsive metal-organic frameworks"

SIMAP laboratory (UMR5266), CNRS

1 Oct. 2017– 31 Dec. 2020 Grenoble, France

- Unraveling microscopic mechanism of light-tunable CO_2 adsorption in photoresponsive metal-organic frameworks (MOFs), relevant for energy efficient CO_2 capture, using atomistic modeling
- Study of strongly bound electron-hole states in metal-organic frameworks (MOFs) and their implications on the optoelectronic potential of MOFs
- Proposing an efficient and accurate *ab initio* QM/MM method for computing optical excitation energies of solvated azobenzene-based photo-switches within GW-BSE formalism
- Side-project: Atomistic modeling of structural evolution of Pt nanoparticle in presence of CO gas
- Mentoring students of Grenoble-INP for semester project in the field of materials modelling

Advisor: Roberta Poloni **Collaborators**: Xavier Blase, Li-Chiang Lin, Claudio Attaccalite, Jing Li

MS thesis researcher

"First-principles investigation of 2D MoS_2-MXene heterostructures as cathodes in Mg-ion batteries"

Jawaharlal Nehru Centre for Advanced Scientific Research (JNCASR)

1 Apr. 2016 - 31 Mar. 2017 Bengaluru, India

- Proposing and modeling MoS_2-MXene heterostructure as a Mg-ion battery cathode to exploit both the fast diffusion channels offered by MoS_2 and large gravimetric capacity offered by MXenes

Supervisor: Umesh Waghmare **Co-supervisor**: Prasentjit Ghosh

PUBLICATIONS

- Maniadi, M., Guesdon, M., Tieriekhov, K., Kshirsagar, A. R., Spirito, D., Martín-García, B., Mercier, N., Arteaga, O., Kepenekian, M., Katan, C. & Abhervé, A. "On the Origin of Circularly Polarized Luminescence in Chiral One-Dimensional (A)$PbBr_3$ Single Crystals". *Chemistry of Materials* **38,** 3496–3509 (2026).
- Even, J., Thebaud, S., **Kshirsagar**, A. R., Pedesseau, L., Zacharias, M. & Katan, C. "Empirical approaches to Frohlich excitonic polarons in polar semiconductors with application to 3D halide perovskites". *arXiv* **2505.07406,** [cond-mat.mtrl–sci] (2025).
- **Kshirsagar**, A. R. & Reichardt, S. "Flipping of electronic spins in $BiFeO_3$ via chiral $d-d$ excitations". *Physical Review B* **112,** L121111 (2025).
- **Kshirsagar**, A. R. & Reichardt, S. "Understanding electronic excited states in $BiFeO_3$ via ab initio calculations and symmetry analysis". *Physical Review B* **110,** 155131 (2024).
- Yetkin, H. A., Dale, P. J., **Kshirsagar**, A. R., Reichardt, S., Latini, A., Ciccioli, A. & Romagnoli, L. "A Combined Theoretical and Experimental Raman Scattering Study of BaZrS3-BaHfS3 Solid Solutions". *ChemPhysChem* **25,** e202400340 (2024).
- **Kshirsagar**, A. R. & Poloni, R. "Assessing the Role of the Kohn-Sham Density in the Calculation of the Low-Lying Bethe-Salpeter Excitation Energies". *The Journal of Physical Chemistry A* **127,** 2618–2627 (2023).
- Ram, A., Maity, K., Marchand, C., Mahmoudi, A., **Kshirsagar**, A. R., Soliman, M., Taniguchi, T., Watanabe, K., Doudin, B., Ouerghi, A., Reichardt, S., O'Connor, I. & Dayen, J.-F. "Reconfigurable Multifunctional van der Waals Ferroelectric Devices and Logic Circuits". *ACS Nano* **17,** 21865–21877 (2023).
- Carnis, J., **Kshirsagar**, A. R., Wu, L., Dupraz, M., Labat, S., Texier, M., Favre, L., Gao, L., Oropeza, F. E., Gazit, N., Almog, E., Campos, A., Micha, J.-S., Hensen, E. J. M., Leake, S. J., Schülli, T. U., Rabkin, E., Thomas, O., Poloni, R., Hofmann, J. P. & Richard, M.-I. "Twin boundary migration in an individual platinum nanocrystal during catalytic CO oxidation". *Nature Communications* **12** (2021).
- **Kshirsagar**, A. R. *Ab initio study of the excited state properties of azobenzene-functionalized photoresponsive MOFs.* PhD thesis (Université Grenoble Alpes, 2021).
- **Kshirsagar**, A. R., Attaccalite, C., Blase, X., Li, J. & Poloni, R. "Bethe–Salpeter Study of the Optical Absorption of trans and cis Azobenzene-Functionalized Metal–Organic Frameworks Using Molecular and Periodic Models". *The Journal of Physical Chemistry C* **125,** 7401–7412 (2021).
- **Kshirsagar**, A. R., Blase, X., Attaccalite, C. & Poloni, R. "Strongly Bound Excitons in Metal–Organic Framework MOF-5: A Many-Body Perturbation Theory Study". *The Journal of Physical Chemistry Letters* **12,** 4045–4051 (2021).
- **Kshirsagar**, A. R., D'Avino, G., Blase, X., Li, J. & Poloni, R. "Accurate Prediction of the S_1 Excitation Energy in Solvated Azobenzene Derivatives via Embedded Orbital-Tuned Bethe-Salpeter Calculations". *Journal of Chemical Theory and Computation* **16,** 2021–2027 (2020).
- Yang, C.-T., **Kshirsagar**, A. R., Eddin, A. C., Lin, L.-C. & Poloni, R. "Tuning Gas Adsorption by Metal Node Blocking in Photoresponsive Metal–Organic Frameworks". *Chemistry – A European Journal* **24,** 15167–15172 (2018).
- **Kshirsagar**, A. R. "First-Principles Investigation of MoS2-MXene Heterostructures as Cathodes in Magnesium Ion Batteries. M.S. Thesis". *Indian Institute of Science Education and Research, Pune* (2017).

TEACHING EXPERIENCE

Initiation to Research, Energy in Chemistry : 27 hours
The undergraduate course comprised of 3h ($\times 2$ batches) lectures on Quantum Chemistry and hands-on training of calculations of formation energies of molecules and analysis of chemical bonding in terms of quantum interference of atomic orbitals.

Cycle pluridisciplinaire d'etudes supérieures (CPES), University of Rennes

2025-26 Rennes, France

In-charge: Xavier Rocquefelte

Tutoring in VASP workshop : 15.5 hours
The workshop comprised of theoretical and hands-on training on density-functional-theory (DFT) calculations using the Vienna Ab-initio Simulation Package (VASP)

University of Rennes and VASP Software GmbH

2025 Rennes, France

In-charge: Xavier Rocquefelte

Mentorship of Masters students: 12 hours
A short research project on determination of molecular structures using density functional theory

Functionalized Advanced Materials Engineering (FAME+) European Masters program, Grenoble-INP, University of Grenoble-Alps

2020-2021 Grenoble, France

In-charge: Roberta Poloni

Mentorship of Masters students: 6 hours
A short research project on optimisation of structure of porous materials using density functional theory

Phelma engineering school, Grenoble-INP, University of Grenoble-Alps

2019-2020 Grenoble, France

In-charge: Roberta Poloni **Co-supervisor**: Lorenzo A. Mariano

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Mentorship of school science club
Conducting science workshops for school students

Dr. Vasantdada Patil Municipal School, K. C. Thackeray Municipal School

2014-2016 Pune, India

In-charge: Shraddha Gargatti, Exciting Science Group

ORGANISATIONAL EXPERIENCE

- Organisation of weekly seminars in Theoretical Inorganic Chemistry Unit at ISCR

2025 Rennes, France

- Assistant for symposium *Halide perovskites for quantum technologies* at MATSUS Spring

April 2025 Sevilla, Spain

- Committee member of LuxDoc, an association of nonpermanent researchers

2022-2023 Luxembourg

- Co-organizer of the annual workshop of department of physics and materials science (DPhyMS)

2022 Luxembourg

- Co-organizer of 'Rencontres des Jeunes Physicien(ne)s' (Young Physicists' Meet)

2017 Grenoble, France

- Part of organizing team of March for Science

2017 Pune, India

- General coordinator of 'Mimamsa', pan-India undergraduate science quiz

2015 Pune, India

SKILLS

Computer skills

Python | Matplotlib | Numpy | Fortran (basics) | Linux | High performance computing | Bash | Numerical methods | QuantumEspresso | Abinit | SIESTA | Yambo | Wannier90 | NWChem | Orca | Office | LaTeX | Inkscape

Scientific skills

Quantum mechanics | Quantum chemistry | Solid state physics | Atomistic modeling | Computational material science | Material characterisation | Density functional theory | Molecular dynamics | Structure optimisation | Perturbation Theory | Bethe-Salpeter Equation | Density functional perturbation Theory | Optical spectroscopy | Vibrational spectroscopy | Electronic excitations | Group theory | Chemical bonding | Data analysis | Connecting experiments to theory

Communication skills

Science communication | Article and report writing | Data visualization | Presentation skills | English: C1 | French: B1 | Hindi: C1 | Marathi: Native

ACHIEVEMENTS

Selected for 'Make Our Planet Great Again' (MOPGA) one-year visiting fellowship program

2025 Campus France, French Government

Research proposal for MSCA postdoctoral fellowship received Seal of Excellence

2026 European Commission

Research fellowship by qualifying National Eligibility Test in Physics with all India rank 78

2016 Council of Scientific and Industrial Research (India)

INSPIRE undergraduate scholarship

2012-17 Department of Science and Technology, India

CONFERENCES

ThéMoSiA: West Pole meeting

Talk: Theoretical description of circular dichroism in solids: new directions

Dec 2025 Nantes, France

Psi-K 2025

Poster: Interplay of charge, spin, and lattice via on-site excitons in multiferroic perovskite

Aug 2025 Lausanne, Switzerland

Chiral Phonons workshop, Johannes Gutenberg University Mainz

Poster: Chirality transfer in hybrid organic-inorganic materials : role of chiral vibrational modes

Jul 2025 — Ingelheim, Germany

Journées Pérovskites Halogénées, CNRS GDR

Poster: Modulation of magnetization in BiFeO3 using circularly polarized light

Mar 2025 — Erquy, France

Young Investigator's Meet on Quantum Condensed Matter Theory, IISER Pune

Invited talk: Modulation of magnetization in $BiFeO_3$ using circularly polarized light

Dec 2024 — Pune, India

International Workshop on 2D Heterostructures (HOWDI)

Poster: Modulation of magnetization in $BiFeO_3$ using circularly polarized light

Oct 2024 — Lipperscheid, Luxembourg

March Meeting of American Physical Society (APS)

Talk: Modulation of magnetization in $BiFeO_3$ using circularly polarized light

March 2024 — Minneapolis, United States of America

International Winter School 2023, Jawaharlal Nehru Centre for Advance Scientific Research

Poster: Modulation of magnetization in $BiFeO_3$ using circularly polarized light

December 2023 — Bengaluru, India

GDR-REST: Rencontres de Spectroscopie Théorique (Theoretical Spectroscopy Meeting)

Talk: Resonant coupling of spin-flip excitations with phonons in $BiFeO_3$

June 2023 — Île d'Oléron, France

German Physical Society (DPG) spring meeting

Talk: Resonant coupling of spin-flip excitations with phonons in $BiFeO_3$

March 2023 — Dresden, Germany

EPW 2022: Summer School on Electron-Phonon Physics from First Principles

Poster: First-principles study of resonant Raman scattering in $BiFeO_3$

June 2022 — Austin, United States of America

JTMS 2020: Journées Théorie, Modélisation et Simulations (organized by French Phys. Soc. and French Chem. Soc.)

Talk: Optical properties of photoresponsive MOFs: assessing fragment models using the BSE/GW formalism

Nov 2020 — Online, France

MOFSIM 2019

Poster: First-principles investigation of photo-responsive metal-organic-frameworks for efficient CO2 capture

Apr 2019 — Ghent, Belgium

Theoretical Spectroscopy Lectures

Poster: First-principles investigation of photo-responsive metal-organic-frameworks for efficient CO2 capture

Oct 2018 — Lausanne, Switzerland

Journée de la Matière Condensée (organized by French Physical Society)

Poster: First-principles investigation of photo-responsive metal-organic-frameworks for efficient CO2 capture

Aug 2018 — Grenoble, France

GDR-REST: Rencontres de Spectroscopie Théorique

Talk: First-principles investigation of photo-responsive metal-organic-frameworks for efficient CO2 capture

May 2018 — Porquerolles, France

INVITED TALKS

Tata Institute of Fundamental Research — Host: Dr. Darshan Joshi

Talk: Chiral Quantum States: Interplay of Spin, Charge and Lattice

Nov 2025 — Hyderabad, India

Institute of Chemical Technology, Mumbai — Host: Dr. Ashwin Mohan

Talk: Light-induced processes in materials : insights from ab initio methods

Jan 2025 — Mumbai, India

Institute of Physics and Chemistry of Materials of Strasbourg (IPCMS) — Host: Prof. Mébarek Alouani

Talk: Light-induced processes in materials : insights from ab initio methods

March 2024 — Strasbourg, France

Institute Charles Gerhardt Montpellier (ICGM) — Host: Dr. Guillaume Maurin

Talk: Light tunable gas adsorption in functionalised metal-organic frameworks: insights from ab initio methods

February 2024 — Montpellier, France

Laboratory Charles Coulomb — Host: Dr. Thibault Sohier

Talk: Modulation of magnetization in $BiFeO_3$ using circularly polarized light

February 2024 — Montpellier, France

Institute of Chemical Sciences of Rennes (ISCR), University of Rennes — Host: Dr. Claudine Katan

Talk: Modulation of magnetization in $BiFeO_3$ using circularly polarized light

February 2023 — Rennes, France

Institute of Electronics, Microelectronics and Nanotechnology (IEMN) — Host: Dr. Christophe Delerue

Talk: Modulation of magnetization in $BiFeO_3$ using circularly polarized light

January 2023 — Lille, France

Indian Institute of Science Education and Research (IISER), Pune — Host: Dr. Prasenjit Ghosh

Talk: Modulation of magnetization in $BiFeO_3$ using circularly polarized light

December 2023 — Pune, India

Institute of Molecules and Materiels of Le Mans — Host: Dr. Mads Weber

Talk: Modulation of magnetization in $BiFeO_3$ using circularly polarized light

November 2023 — Le Mans, France

University of Liège — Host: Prof. Matthieu Verstraete

Talk: Understanding excitons in $BiFeO_3$

April 2023 — Liège, Belgium

Interdisciplinary Center of Nanoscience of Marseille — Host: Dr. Claudio Attaccalite

Talk: Ab initio study of photo-responsive metal-organic frameworks for an efficient carbon capture

January 2020 — Marseille, France

PEER-REVIEW ACTIVITY

Physical Review Letters, Physical Review B, Physical Review Materials (American Physical Society)

2024-present

OTHER AFFILIATIONS

- Member of American Physical Society, APS (2023-2024)
- Member of German Physical Society, DPG (2022-2024)
- Member of European Theoretical Spectroscopy Facility, ETSF (2022-present)

REFEREES

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